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portafolio

ROXANA FIORELLA GUILLEN HURTADO

Postgraduate student in Urban Spatial Science, combining a background in architecture and urbanism with a data-driven approach to tackle urban challenges. Skilled in spatial analysis, sustainable planning, and participatory design, with hands-on experience working with local governments.

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You can access an interactive version of my portfolio at:

fiorellaguillen.github.io/portfolio/

education

MSc Urban Spatial Science
Class of 2025 | UCL - The Bartlett Centre for Advanced Spatial Analysis

Bachelor in Architecture
Class of 2021 | PUCP - Faculty of Architecture and Urbanism

Movetia Workshop
2022 | PUCP & Bern University of Applied Sciences (BFH)

Study-Abroad Semester | MSc. ITECH
2018-19 | University of Stuttgart

Adaptive Transport Machine Workshop
2017 | Architectural Association Visiting School

awards & accreditations

Registered Architect
Peruvian Association of Architects

Bicentennial Generation Scholarship
Peruvian Government

International Grand Prize for Sustainable Urban Design
V Edition of the Architectural and Urban Sustainable Design Award - ADUS 2021

Circular economy and sustainability
International Biennial of Architecture and Urbanism - CAP, Peru

2022 & 2018 Excellence award
Faculty of Architecture and Urbanism PUCP

skills

Programming

Python | R | SQL | Netlogo | Google Earth Engine

Design Software

Autocad | Archicad | Sketchup | Rhinoceros | Grasshopper

Illustrator | Photoshop | InDesign | Lumion | Premiere

Languages

Spanish - native | English - C1 | German - B1

URBAN DESIGN

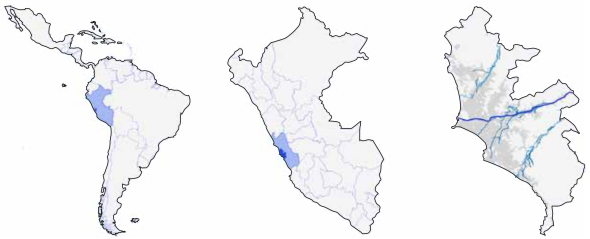
Master Plan for the Rimac River

LIMA, PERU

The **Rímac River Landscape Project** (PEPRR) is one of Lima's most ambitious urban initiatives, including 66 key interventions, from local parks and plazas to major avenues and emblematic public spaces. To guide the project, a comprehensive master plan was developed, integrating urban, social, transport, and heritage considerations to identify strategic opportunities for sustainable transformation.

The core objective was to **restore the Rímac River as a vital ecological and urban axis** through a connected system of inclusive, sustainable public spaces and a mobility network focused on pedestrians and public transport. The project includes 28 safe and resilient public spaces that address service access gaps, while also preserving Lima's rich natural and cultural heritage, spanning pre-Hispanic to contemporary periods, and boosting tourism and local economic activity.

My contribution involved direct participation in **stakeholder and community meetings, helping shape design guidelines, identifying strategic intervention opportunities, developing graphics and renders, and designing sustainable mobility guidelines.**



Perú *country*

Lima *region*

Lima *province*

Master Plan

Project indicators

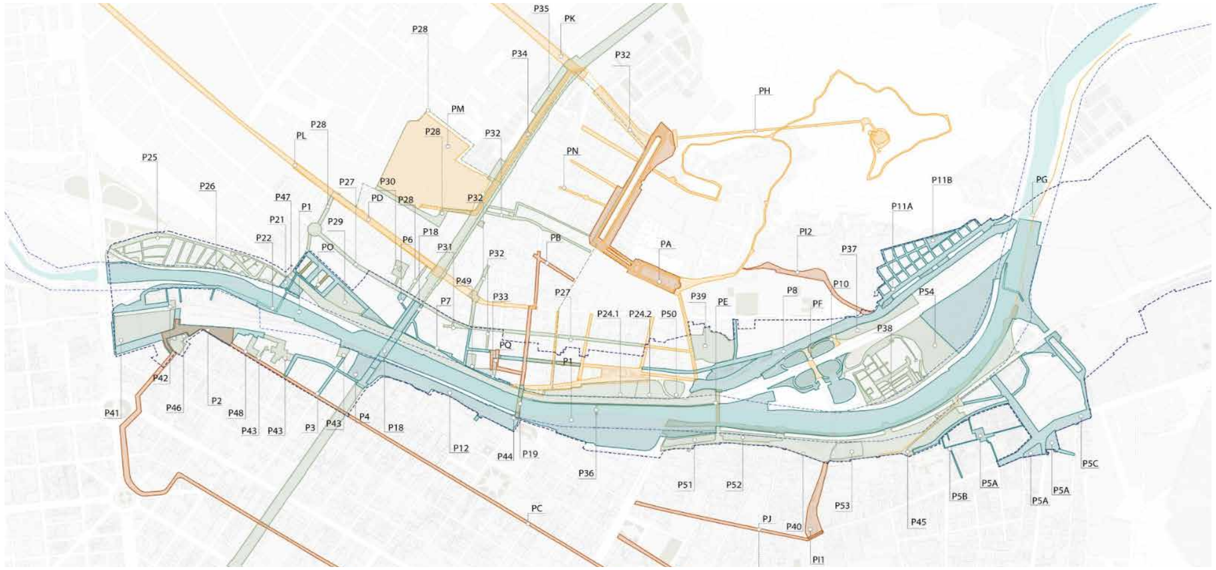
17 km	**20** km	**2** un	**4** km	**9** km	**5076** ml	**31.2** ha	**3** un	**1454** ml	**2** un	**7530** ml	**9** ha
Redesigned avenues and streets	*Cycle lanes (shared and segregated)*	*Tram routes integrated*	*Railway (section within the CHL)*	*Tree-lined paths*	*Revegetated riverbanks*	*Riparian forest*	*Restored historic canals*	*Bioremediation zones*	*Water control and improvement centers*	*Riverine defense*	*Revegetated islets*

Potential interventions identified

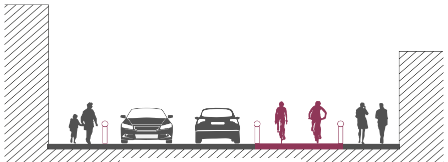
66 Interventions

1 + **14** + **7** + **1** + **24** + **7** + **10** + **2**

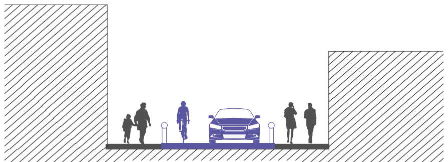
1	14	7	1	24	7	10	2
Rimac River renaturalization	*Park projects*	*Squares projects*	*Arborization projects*	*Street redesign projects*	*Bridges projects*	*Heritage buildings restoration*	*New buildings projects*



Bike lane network proposal



Segregated bike lanes



Shared bike lanes

General renders of the Rimac River Masterplan

1. View of the Rimac River redesign, including the new Chabuca Granda Promenade and the Estancos de Sal park.

2. View of the project's impact on the Rimac district, including the revegetation of its streets and San Cristobal hill.

Connecting the master plan to the city's networks

Participatory design and stakeholder meetings






URBAN DESIGN

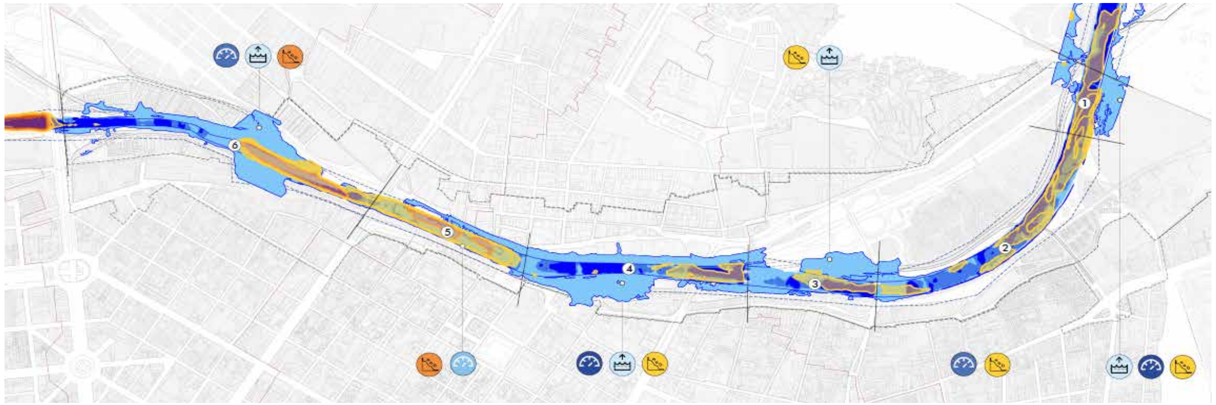
Hydrogeo-morphological Manual for the Rimac River

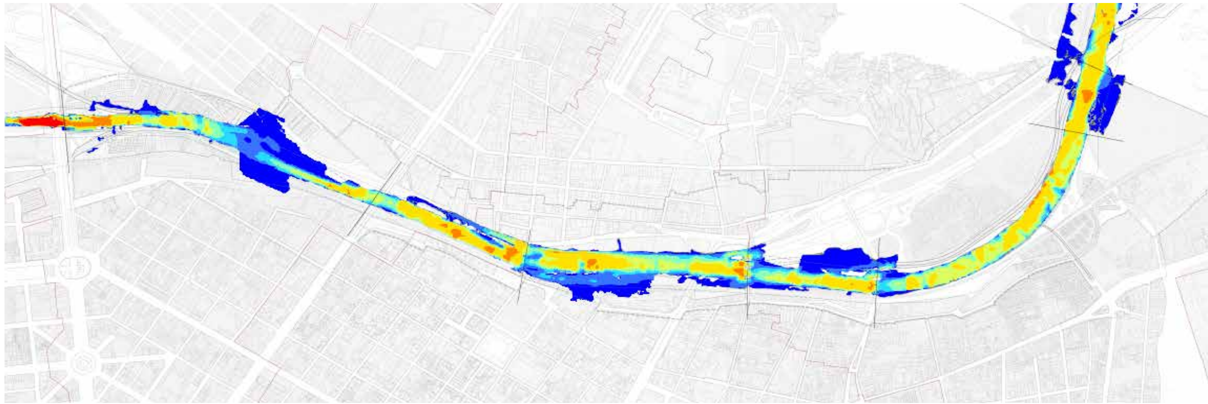
LIMA, PERU

Developed as part of the **Rímac River Landscape Project**, this study serves as a foundational tool for proposing interventions along the riverbed and banks of the Rímac River. It offers an initial approach to formulating strategies and general guidelines that strengthen the relationship between the Rímac River and the Historic Center of Lima (CHL).

The study includes a comprehensive analysis from historical, hydrological, hydraulic, geomorphological, and landscape perspectives.

Based on this diagnosis, a set of **guidelines** was proposed **for the management and planning of urban-riverine spaces**. These aim to mitigate risks related to erosion and flooding, while restoring the Rímac River's role as a vital hydrological, environmental, and landscape resource.

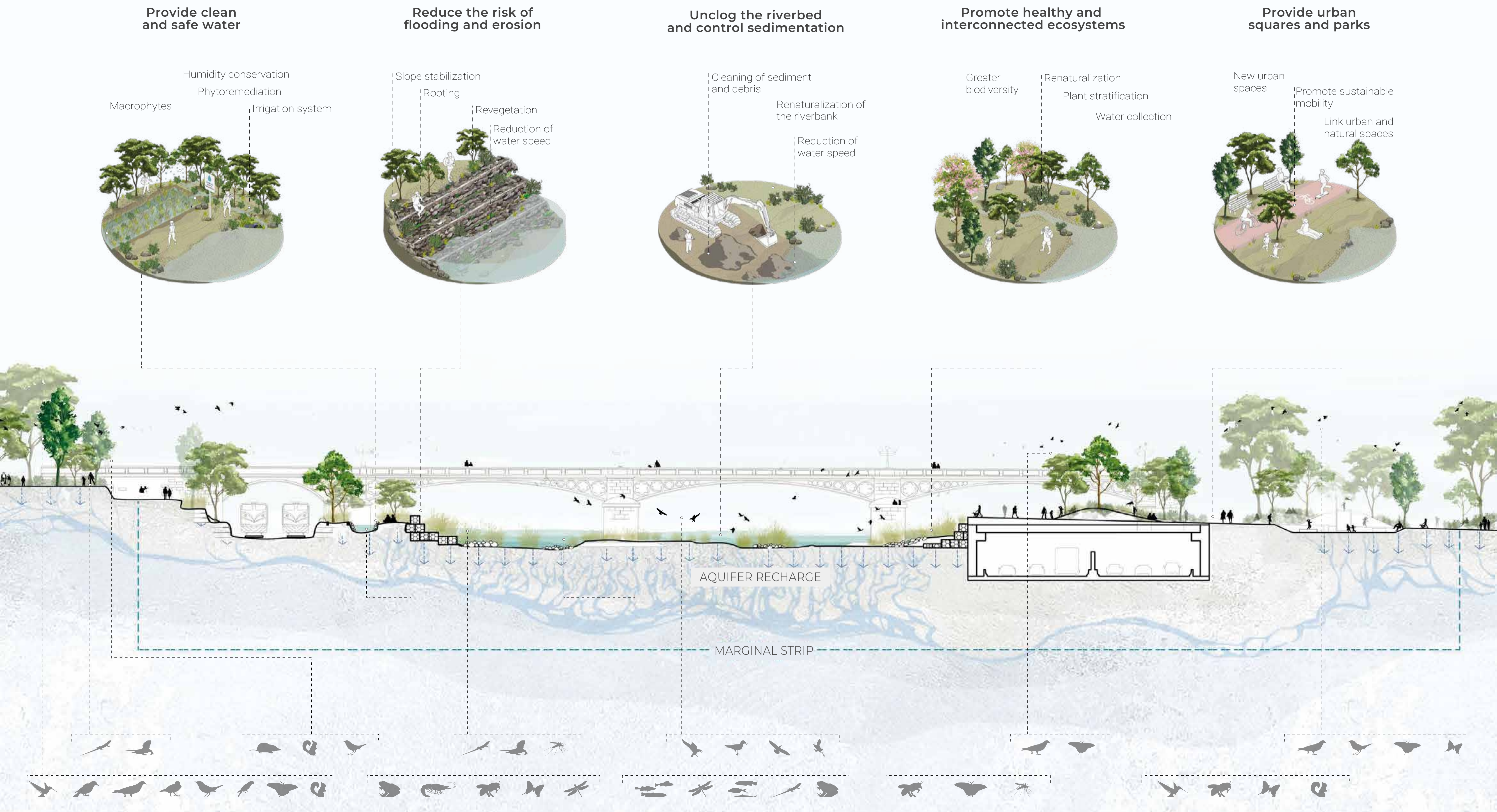




Hydrological maps of the Rimac River (from top to bottom)

1. Hydraulic variable diagnostics by study segments
2. Maximum flow velocities of the Rímac River
3. Flow depth during flood season of the Rímac River

Summary of the suggested interventions



URBAN DESIGN

Ayabaca Street

Urban Mobility and Public Space Impro-vement

LIMA, PERU

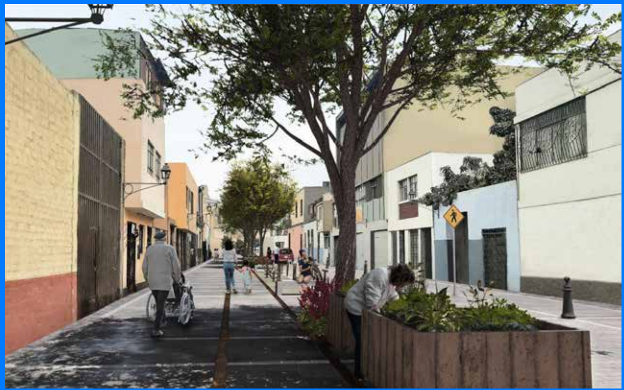
This project focused on **enhancing sustainable urban mobility and revitalizing public space along Ayabaca street,** within the district of Rímac in Lima. The intervention redefines the street as a space for people rather than cars, connecting it with nearby landmarks such as the Barraganes square and the adjacent church, while reinforcing local heritage elements like old tram lines and religious routes.

The proposal promotes walkability, expands green areas, and introduces new public uses such as playgrounds, sports zones, and community gardens. Sidewalks and streets are reconfigured to respond to current mobility needs, and the public realm is upgraded with urban furniture, lighting, and vegetation.

Integrated into the broader urban and heritage projects of the PEPRR, this intervention **aims to reactivate neighborhood life, enhance cultural value, and improve the overall quality of the urban environment.**

14 units	**204** m2	**9** units
New trees	*Green areas*	*Communal gardens*
79 m2	**24** m2	**315** m2
Kids playgrounds	*Sports zone*	*Rest & stay areas*

Renders of the project and participatory process photos



Project's plan and main strategies

CONNECT

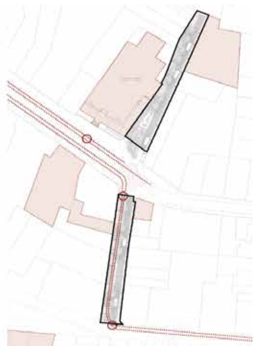
Safe pedestrian-priority circulation, with a green axis of planters and trees

REACTIVATE

Through seating areas, recreational spaces, playgrounds and sport zones

REVALUE

Historic tram lines and the urban grids of Cercado de Lima and Rímac



URBAN DESIGN

Alternative urban model for Bajo Belen

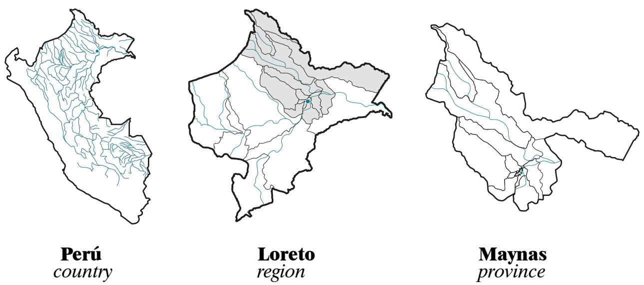
IQUITOS, PERU

This research, developed as my bachelor degree's thesis, proposes a **sustainable and adaptive urban model to address the challenges faced by the flood-prone neighborhood of Bajo Belen**, a cultural and economic hub in the Peruvian Amazon.

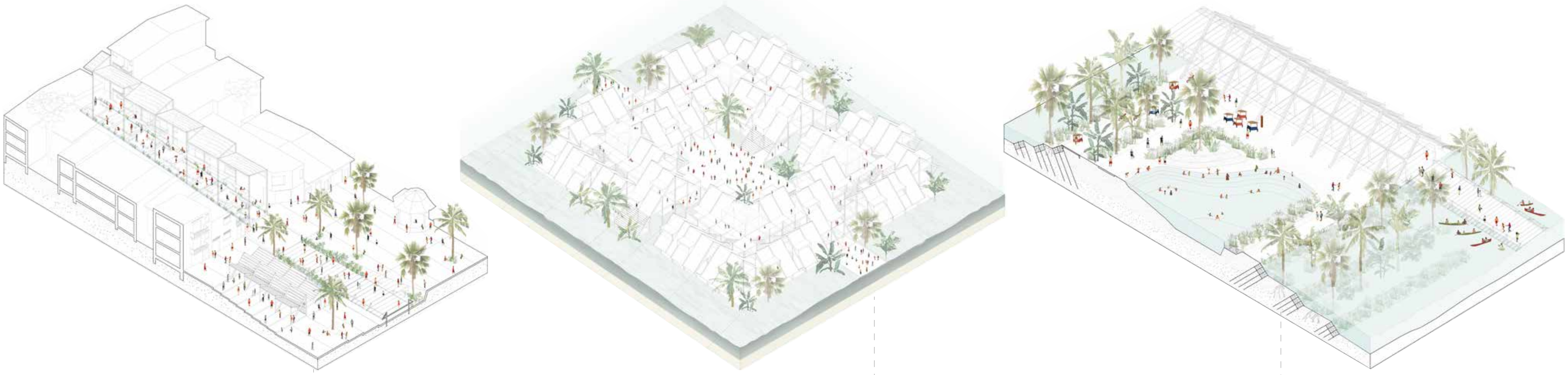
Amid threats of relocation and the loss of territorial identity, the model aims to preserve the essence of Bajo Belen while avoiding current issues such as disconnection, overcrowding, and vulnerability to flooding.

The proposal consists of a **two-scale intervention**: first, a master plan with three key intervention areas: connector area, residential area, and a mitigation & landscape border, and transversal axes to integrate them; second, an urban and architectural design framework for elements like commerce, roads, housing, communal spaces, docks, and pools, tailored to local techniques, climate, and culture.

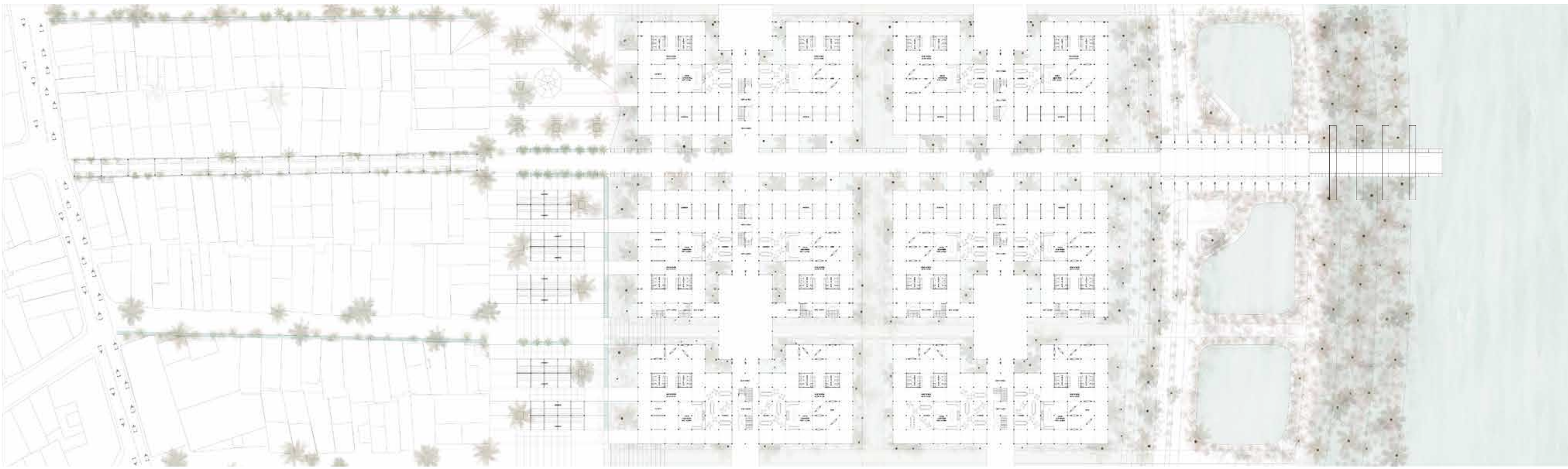
This intervention demonstrates the potential for **a new way of inhabiting the river**, preserving social, economic, and environmental connections while ensuring a safe and dignified future for the community.



Plan and isometric views of the three key intervention areas

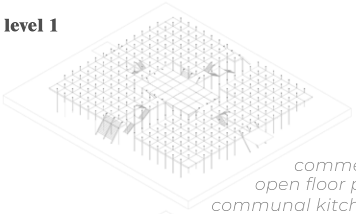


Connector area | **Residential area** | **Mitigation & Landscape border**



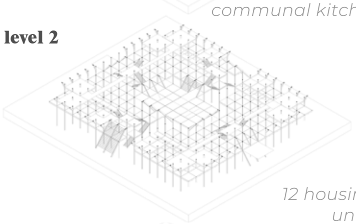
Proposed housing typologies

level 1



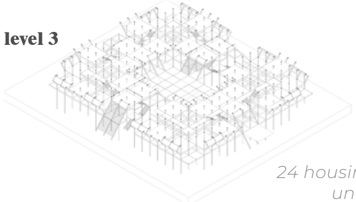
commerce
open floor plan
communal kitchens

level 2



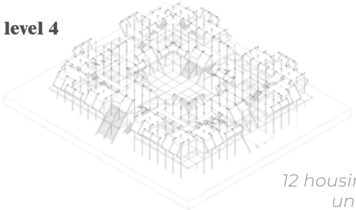
12 housing units

level 3



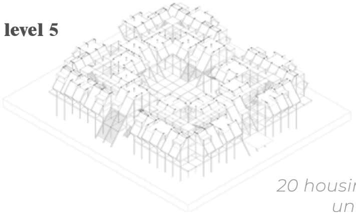
24 housing units

level 4



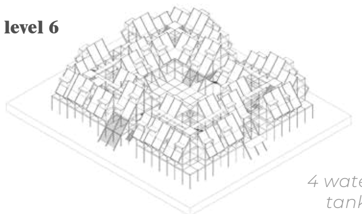
12 housing units

level 5



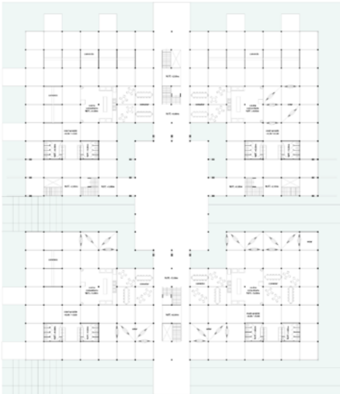
20 housing units

level 6

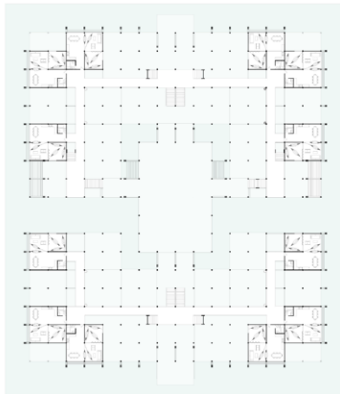


4 water tanks

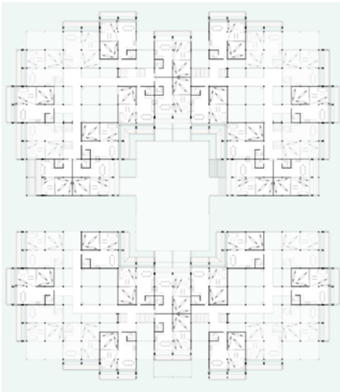
level 1



level 2



level 3



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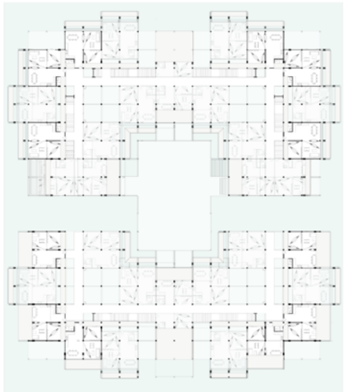

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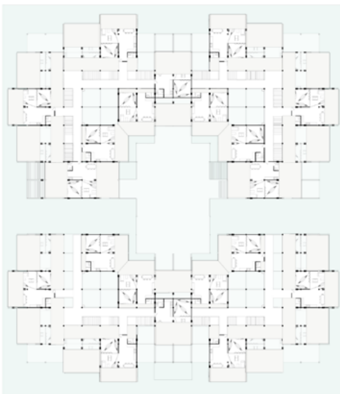
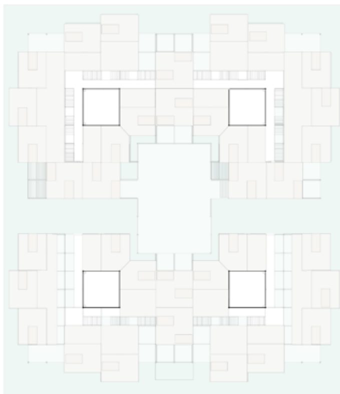


level 6



Illustrations of the project's main areas

URBAN DESIGN

Santa Rosa Market Square

LIMA, PERU

This project seeks to **reclaim and redesign the public space around the Santa Rosa Flower Market,** located along the Tacna Avenue corridor in Lima's Historic Center. The intervention reimagines this key urban node as a **vibrant plaza that prioritizes pedestrians, reorganizes vehicular flow, and enhances connections to surrounding heritage and mobility networks**.

The design draws inspiration from the historical Qhapaq Ñan route and incorporates local landmarks into a renewed civic experience. Through a phased approach, the first stage reclaims underused street areas to create a new plaza adjacent to the market, providing shaded resting spaces, green elements, and improved transit access. New pavement treatments, tree planting, street furniture, and interpretive features enrich the spatial quality and cultural value of the area.

In the second phase, nearby Espinoza Street will be reconfigured with a single-level platform that balances vehicular access and pedestrian comfort. Wider sidewalks, improved crossings, and additional greenery will contribute to a safer, more inclusive environment.

26 units
New trees

355 m2
New green areas

1230 m2
Rest & stay areas

2870 m2
New public space

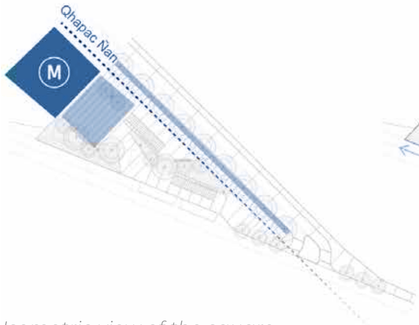
1 unit
Redesigned bus stop

150 ml
Pedestrian priority road

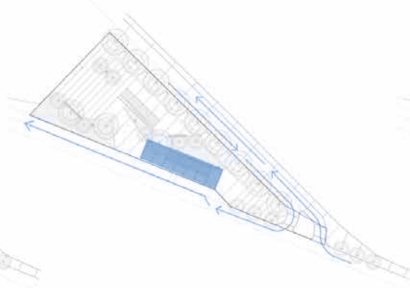
Renders of the project

Strategies of the project

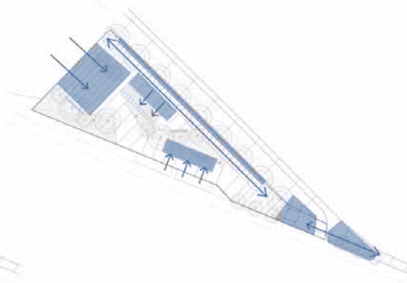
REVALUE
the historic Qhapac Ñan route and the Santa Rosa Flower Market



REORGANIZE
vehicular flows and pedestrian circulation, including a bus stop



REACTIVATE
resting areas by introducing new gathering spaces and a greater variety of uses



Isometric view of the square

URBAN DESIGN

Santa Rosa Park Urban Revitalization

LIMA, PERU

This project reimagines Santa Rosa Park and its surrounding streets - an area of historical, urban, and social significance in Lima's Historic Center, where the current car-centric design has diminished its heritage value and limited pedestrian and cyclist access.

The proposal prioritizes **sustainable and active mobility as a strategy for revitalization**. Key interventions include diverting vehicular traffic away from Jr. Conde de Superunda, creating a shared street with widened sidewalks, and improving pedestrian connectivity between the park and the Basilica.

A new bikeway will link the area to the Rímac River, while adjacent blocks will be upgraded with greenery and street furniture to foster local community life. The redesign of the park includes a unified layout that enhances the central plaza, integrates the Basilica's presence with landscaped buffers, and highlights the historic Monserrate canal through special paving.

By integrating urban design with heritage conservation and mobility strategies, the project aims to **strengthen the identity of this emblematic space and improve quality of life for its residents**.

General plan of the project

Current situation of the intervention area

Environmental benefits of the project

Project indicators

Santa Rosa Park indicators

6023 m2 — *Redesigned public space*

50 units — *New trees*

2600 m2 — *New green areas*

Santa Rosa streets indicators

2280 m2 — *Redesigned public space*

14 units — *New trees*

29 units — *New street furniture*

Renders of the park's redesign

Park redesign strategies

CONECTIVITY

Single platform connects the square with the street and basilica

GREEN AREAS

Integrate green space across three planters and expand vegetation

TREES

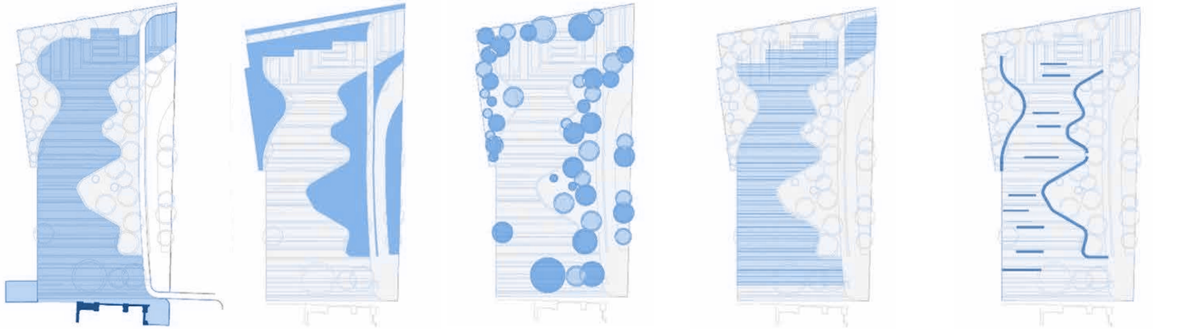
Include native species with environmental benefits

PAVING

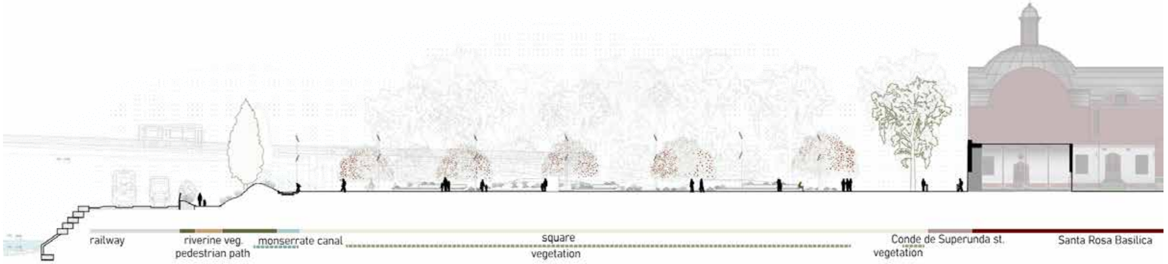
Paving rhythm that relates to the church and materials according to master plan

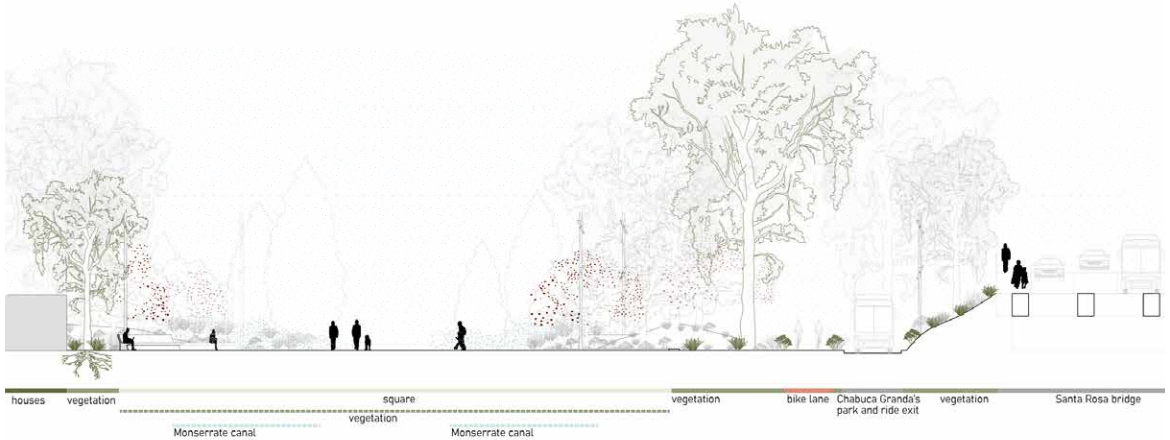
FURNITURE

Benches along the sides of the square and floor details to guide flows



Park sections

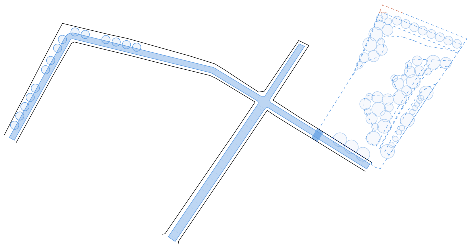




Strategies for the streets project

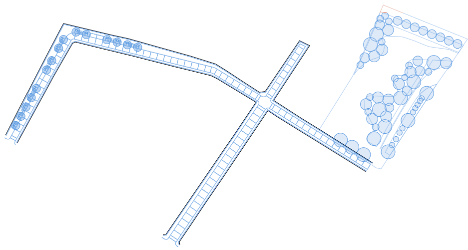
CONECTIVITY

Single platform will give priority to pedestrians and connect with the square



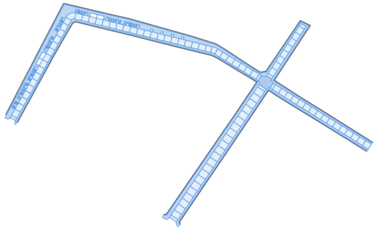
VEGETATION

Add new trees of native spaces that will improve shade in urban spaces



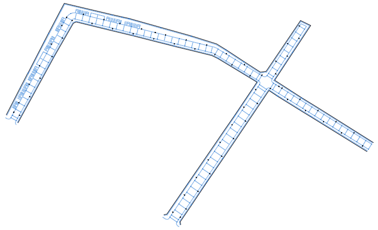
PAVING

Paving materials according to master plan and related to contiguous projects



FURNITURE

Add street furniture to improve lightning and provide resting spaces



Changes on the streets sections

Renders of the streets redesign

URBAN DESIGN

Tacna Avenue Redesign

LIMA, PERU

This ongoing project, developed as part of the Rímac River Special Landscape Project (PEPRR), reimagines **Tacna and Prolongacion Tacna Avenues** as one of the main gateways to Lima's Historic Center and the Rímac district. Building on the central segregated bus corridor proposed by ATU, the intervention enhances the plan by **introducing key urban design strategies to elevate the spatial quality, comfort, and experience of pedestrians and transit users** alike.

The new street section includes wide sidewalks, dedicated bus lanes with optimized stop locations based on pedestrian flow studies, and a fully segregated bikeway to encourage sustainable mobility. The layout also incorporates superblocks that restrict left turns and reduce vehicular conflict, allowing the creation of pedestrian-priority streets in adjacent blocks.

In addition to its mobility function, the **project envisions Tacna as a green corridor**. Vegetation, urban furniture, and resting areas are introduced along the avenue. Three adjacent parks within the Rímac district are also being redesigned and integrated into the corridor through participatory processes.

732 units
Trees

24180 m2
Pedestrian area

3 units
Parks integrated

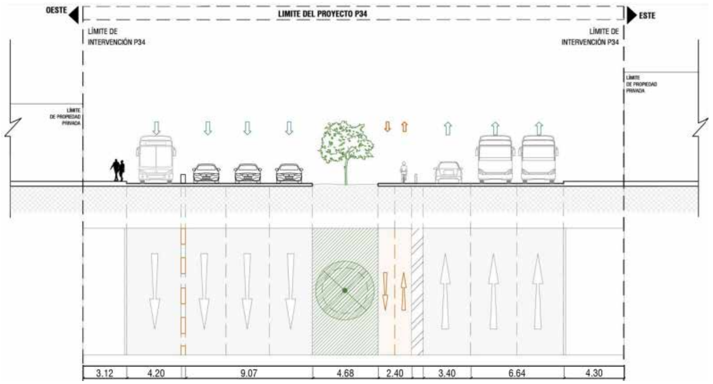
8.2 ha
Total area redesigned

4380 ml
Cycle lane

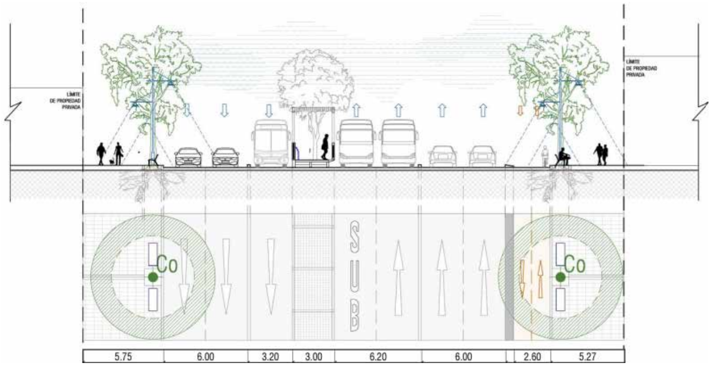
1872 m2
Bus stops infrastructure

Redesign proposal for Tacna Avenue

Current section of Tacna Avenue



Proposed section for the Tacna Avenue



1. Render of Prolongacion Tacna Avenue: park integrated to the design
2. Render of Tacna Avenue: Bike lane and bus stop
3. Current situation of Prolongacion Tacna Avenue
4. Current situation of Tacna Avenue

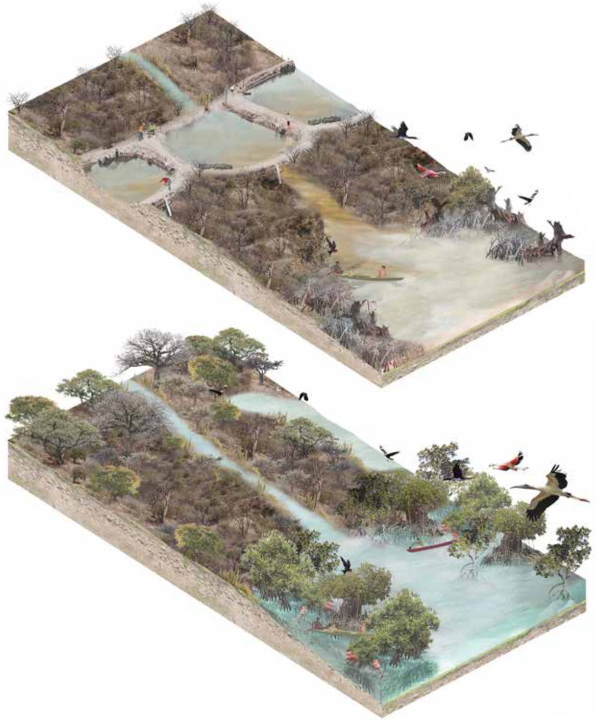
URBAN DESIGN

Design competitions
Visualisations

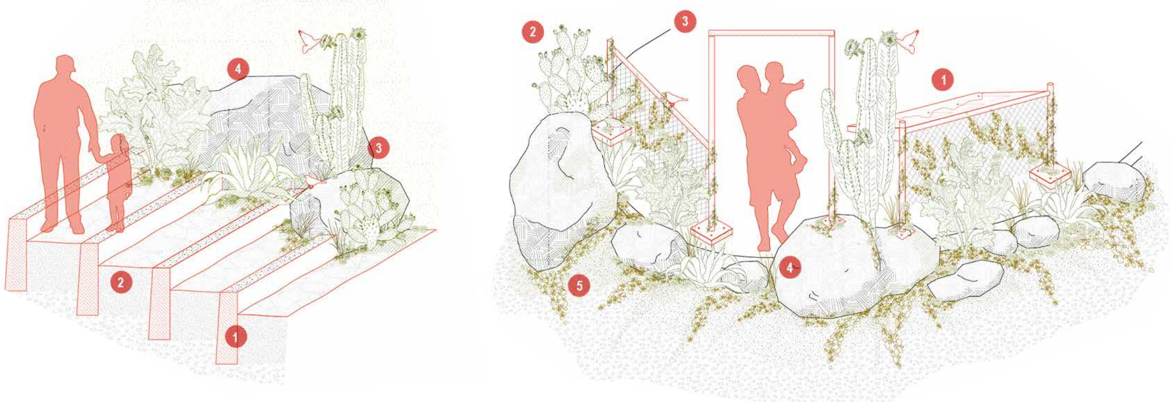
PERU, BOGOTA, MEXICO

As part of my freelance projects, I have developed **graphic proposals for various international competitions**, demonstrating **proficiency in a wide range of visual styles**, from realistic to technical, using rendering software and photomontage techniques with Adobe Creative Suite.

Before and after isometric view of proposal for "Manglares de Tumbes" competition

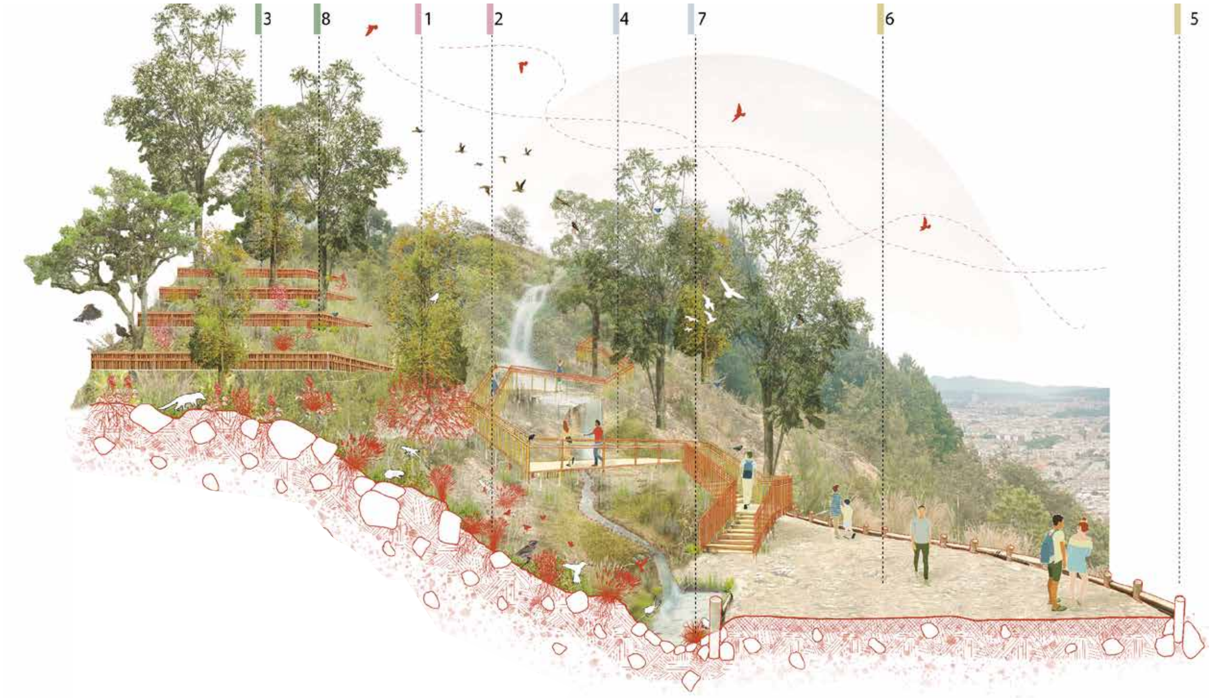


Landscape detail drawings for “Laderas in San Juan de Lurigancho” competition



Urban masterplan for “Mextropoli 2023” competition

Photomontage of proposed viewpoint for “Alternativas de Uso Publico - Cerros orientales” competition



URBAN DATA SCIENCE

Cycling Accident Severity in London

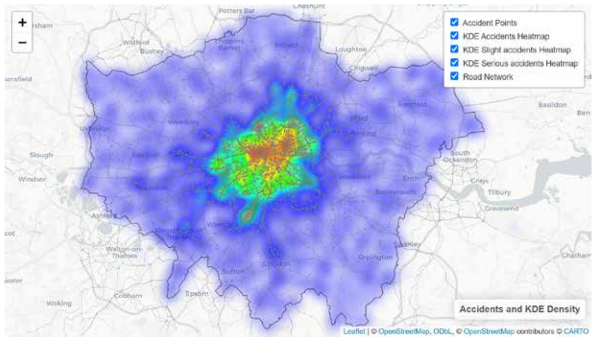
LONDON, UK

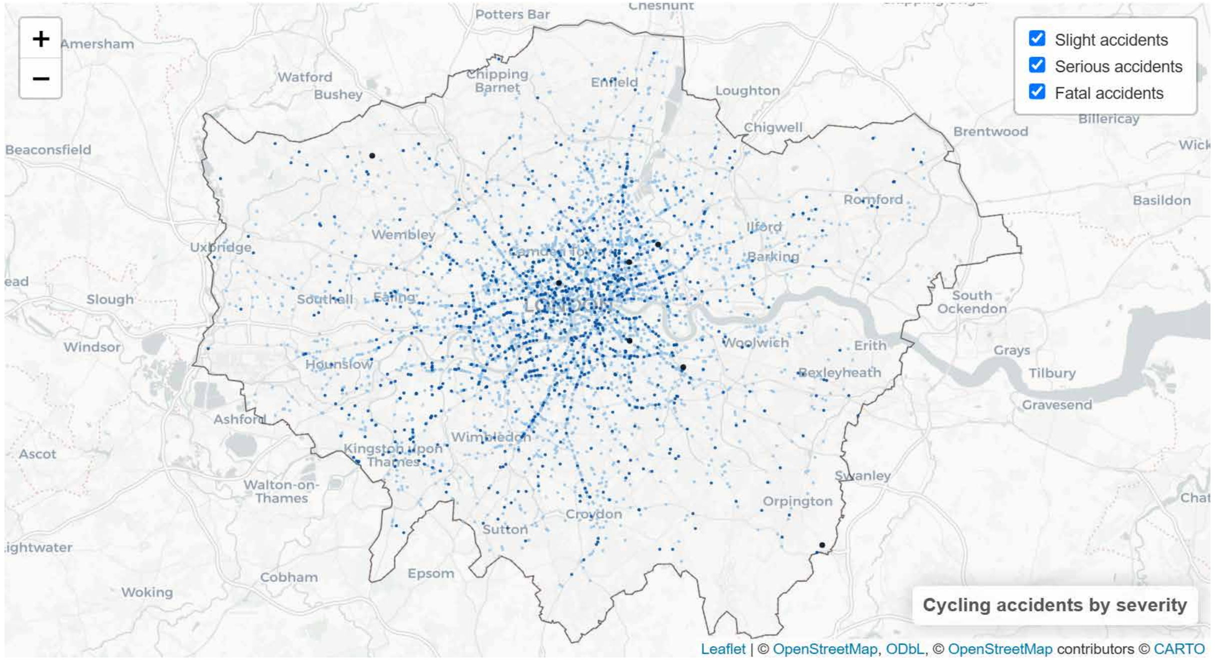
This study examines the **spatial distribution of cycling accidents in London, categorized by severity**. Using Kernel Density Estimation (KDE), it visualizes **high-risk zones and introduces a ranking tool to identify the most dangerous intersections for cyclists**, supporting evidence-based traffic safety prioritization.

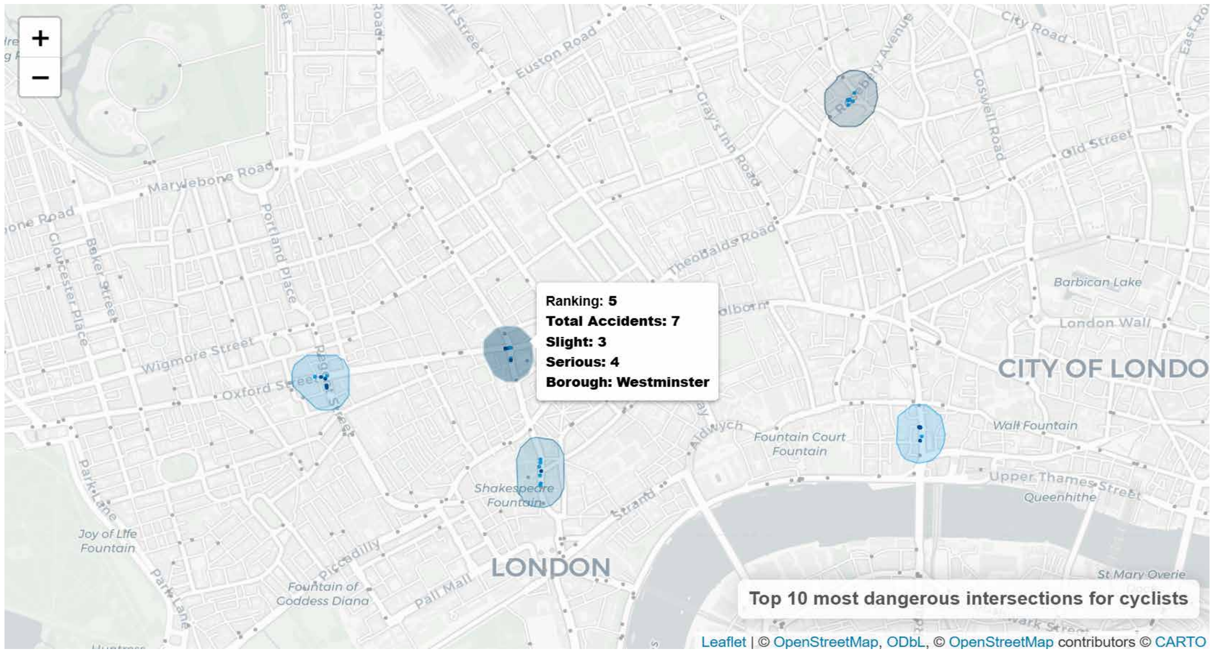
To explore the key factors influencing accident severity, a regression analysis was conducted. The results reveal an inverse correlation between severe accidents and rush hour periods, indicating that time of day significantly shapes accident severity patterns.

An interactive version of the analysis can be found at:

https://fiorellaguillen.github.io/portfolio/projects/urban-data-science/cycling_accidents.html







URBAN DATA SCIENCE

London Bikesharing System

Agent-Based Model

CITY OF LONDON, LONDON, UK

This agent-based model, developed in NetLogo, simulates bikesharing dynamics in the City of London to assess how **access to real-time information via a mobile app affects user behavior and system performance.** The model uses spatially explicit data to compare informed and uninformed users in terms of trip success, time efficiency, and rerouting behavior.

Through this project, I integrated urban mobility data, behavioral modeling, and simulation techniques to support data-driven transport planning.

A simulation of the model can be found at:

https://fiorellaguillen.github.io/portfolio/projects/urban-data-science/bikesharing.html

URBAN DATA SCIENCE

White-crane Habitat Moni-toring App

EAST ASIA

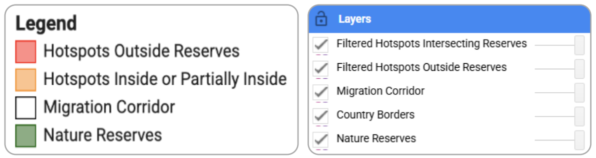
This interactive tool was developed to **identify and map key habitat hotspots for the migratory White-naped Crane across East Asia**, with a particular focus on **areas outside formal nature reserves**.

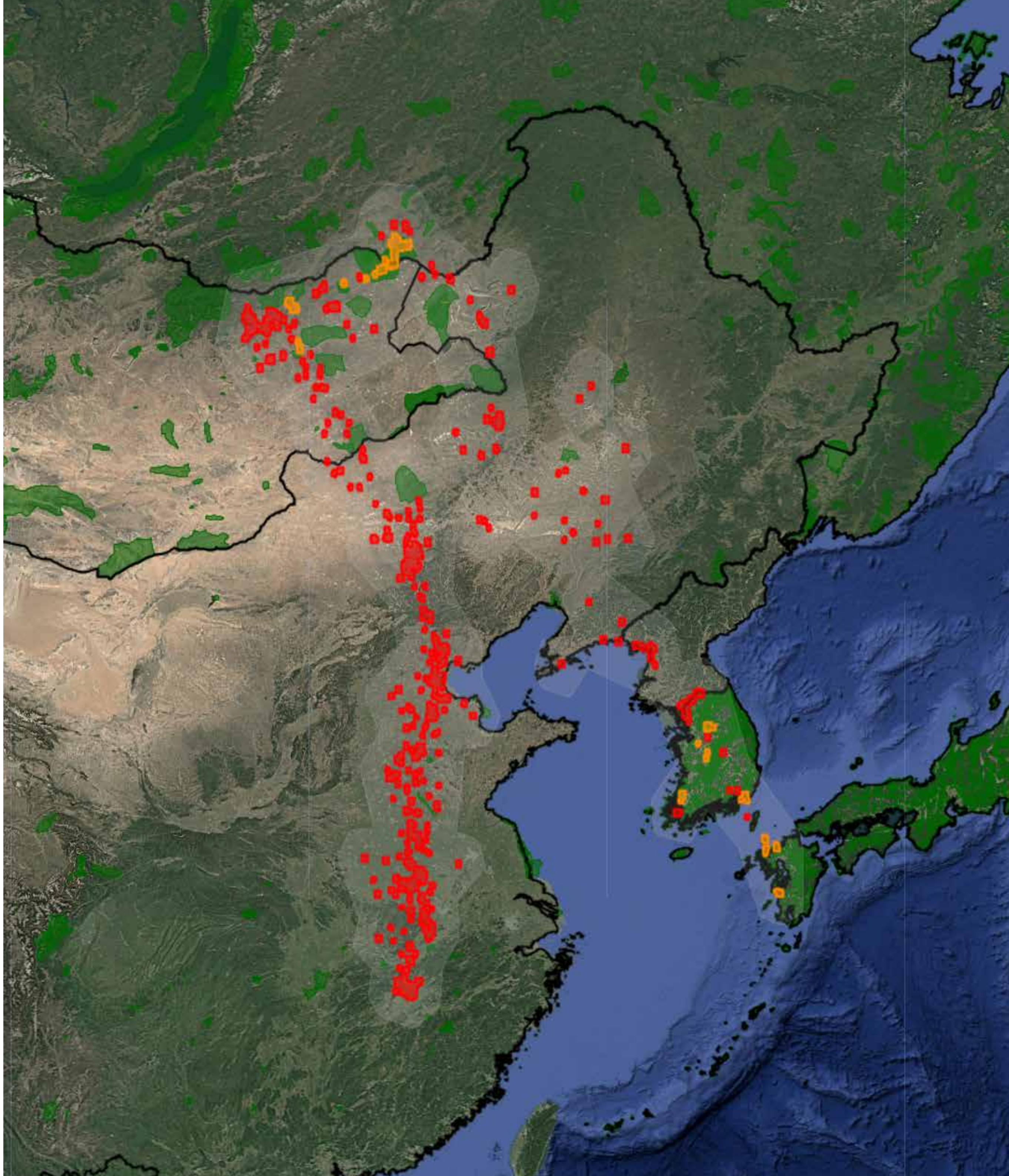
By integrating GPS tracking data with environmental indicators extracted from satellite imagery - such as vegetation cover, tempe-rature, pollution levels, and water availability - it **assesses the ecological condition of these habitats and provides strong, data-driven support for off-reserve conservation efforts**.

Built on Google Earth Engine, the application features an intuitive interface that allows users to define custom regions along the migration corridor and explore seasonal and spatial trends in crane density and habitat quality.

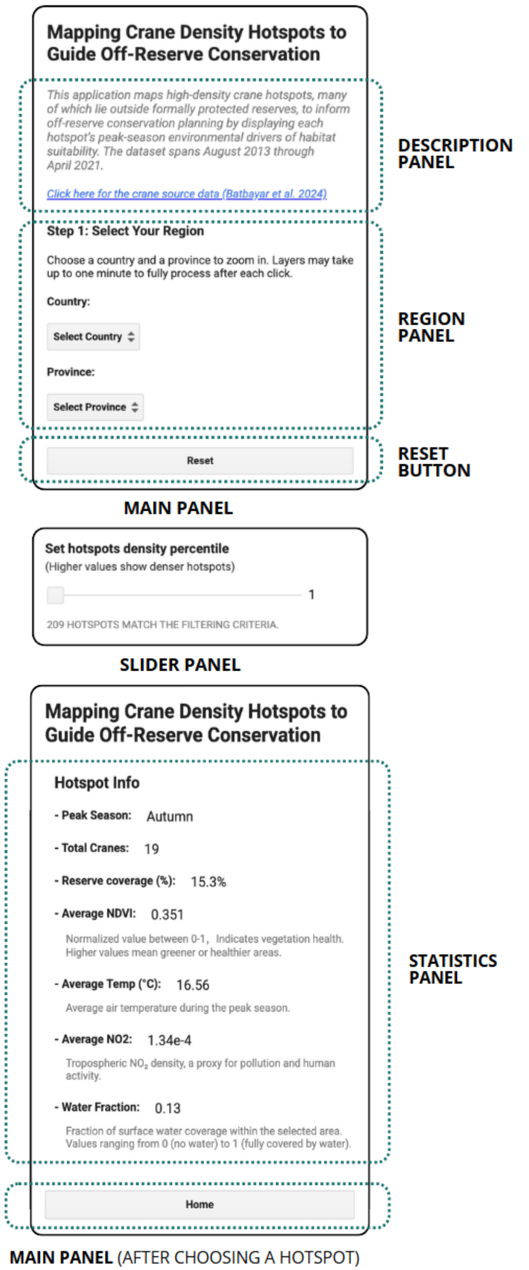
More information on the app and how it was build is available at:
https://lzq407.github.io/CASA0025_Project_Template/

The app is available at the following link:
https://casa25gw.projects.earthengine.app/view/cranes-in-transit

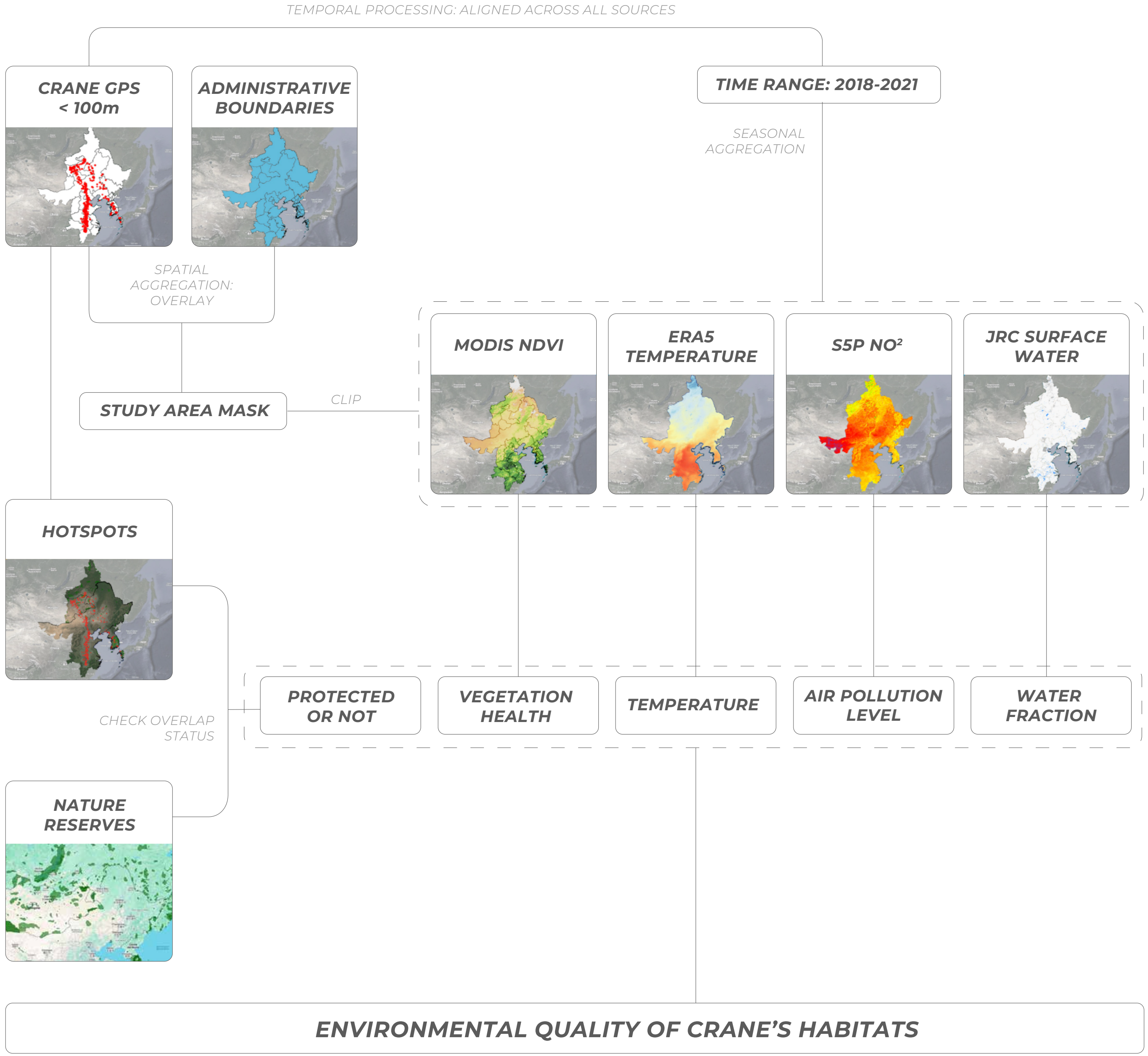




UI Components



Processing structure



PARAMETRIC DESIGN

Parametric pavillion:

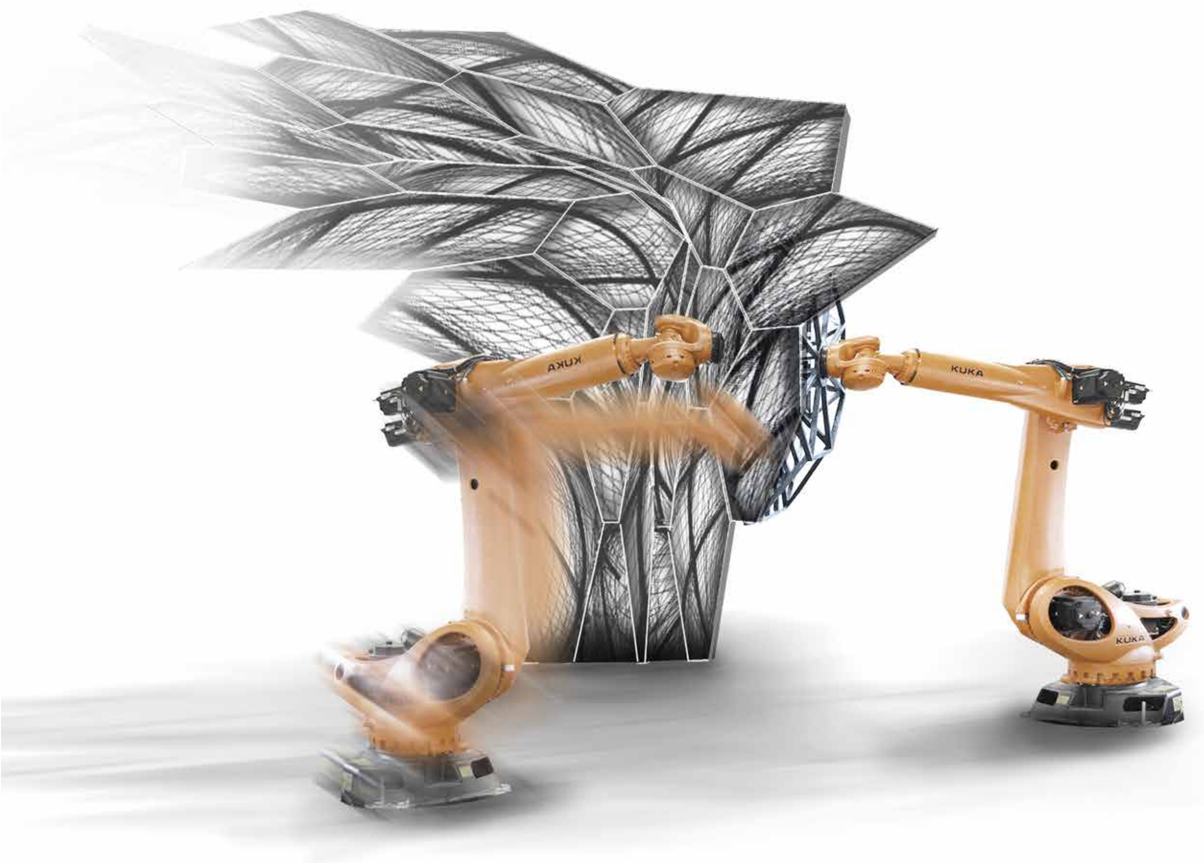
Robotic construction inspired by weaver ants

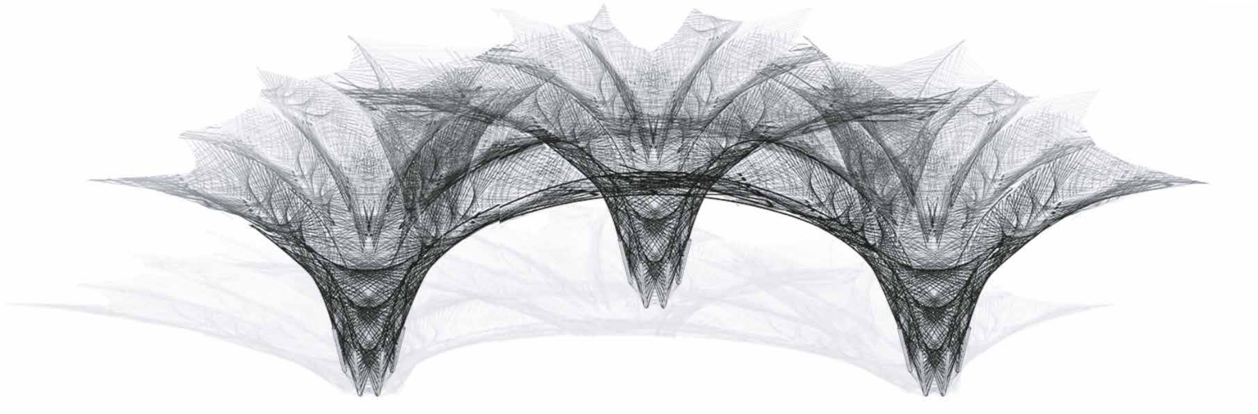
STUTTGART, GERMANY

During my semester in the ITECH program, focused on computational design and digital fabrication for innovative construction processes, I participated in the **design and construction of a research pavilion**, a core component of the curriculum.

Our team proposed a novel construction method inspired by weaver ants, featuring full automation through two robots controlled by a Grasshopper algorithm. The first robot dynamically positioned a scaffold that acted as a mold for the structure, while the second robot wove the structural material around it in real time.

The project involved **biomimetical research, parametric design, and the prototyping** of our proposal.





PARAMETRIC DESIGN

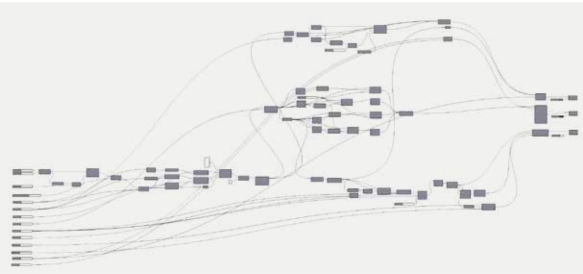
Parametric design instructor

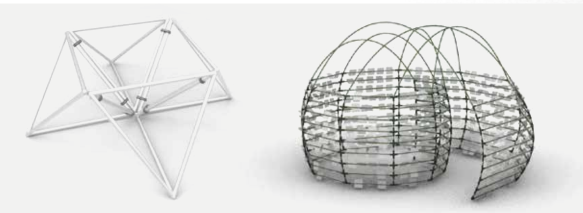
LIMA, PERU

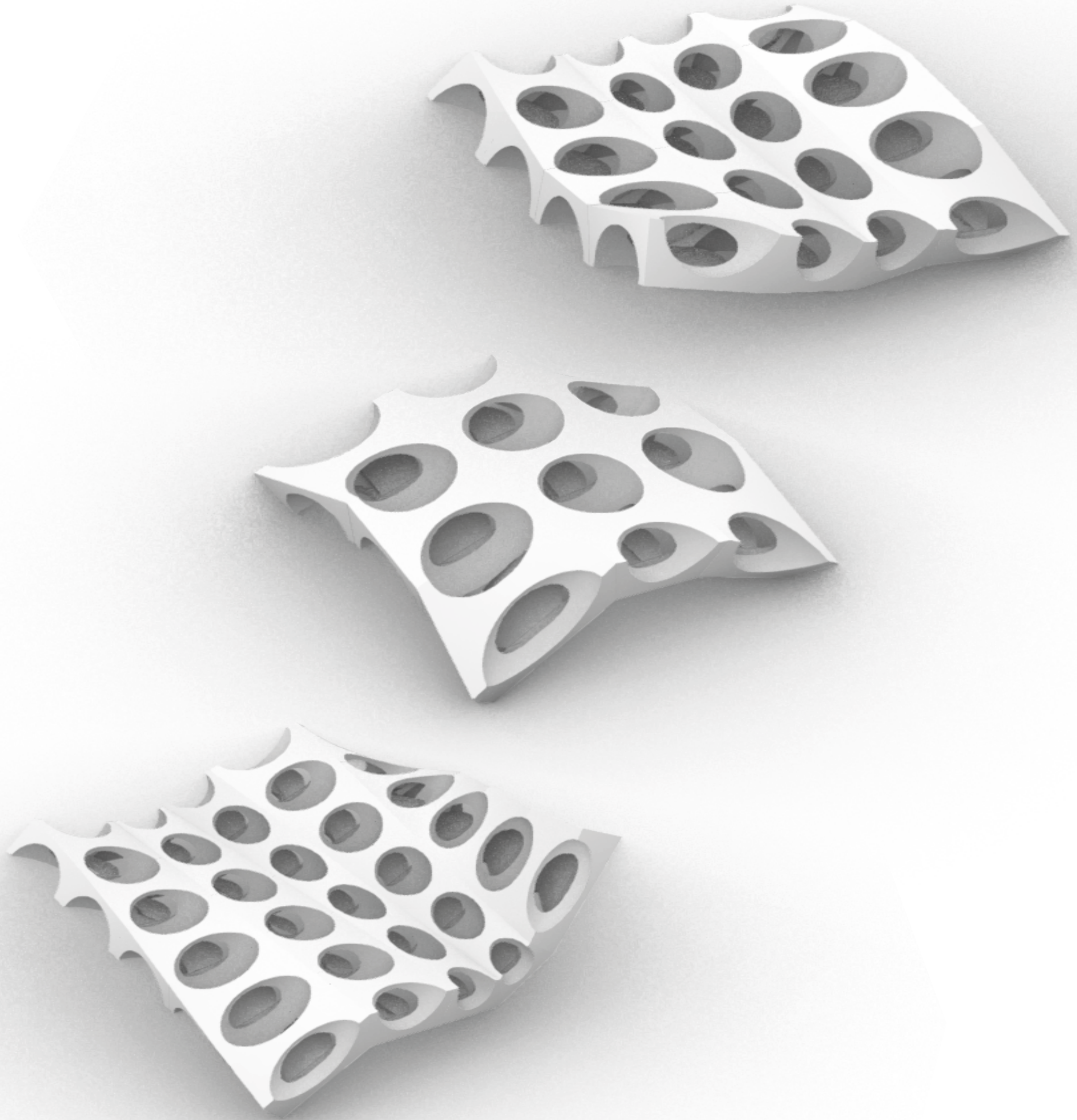
For the past six years, I have been teaching intensive courses on parametric design using Rhinoceros and Grasshopper. These workshops combine theoretical and hands-on sessions, covering key topics such as data structures, and the creation and manipulation of surfaces, meshes, breps, and subDs.

Throughout the course, students gain a solid understanding of parametric logic and geometry management. They learn how to **design and control complex forms algorithmically, adapting them to different constraints and design goals**.

By the end of the course, participants are able to model fully parametric architectural structures, export them to Rhinoceros, and produce high-quality visualizations to communicate their projects effectively.







PARAMETRIC DESIGN

Biomimetic kerfing

Robotic construction inspired by weaver ants

STUTTGART, GERMANY

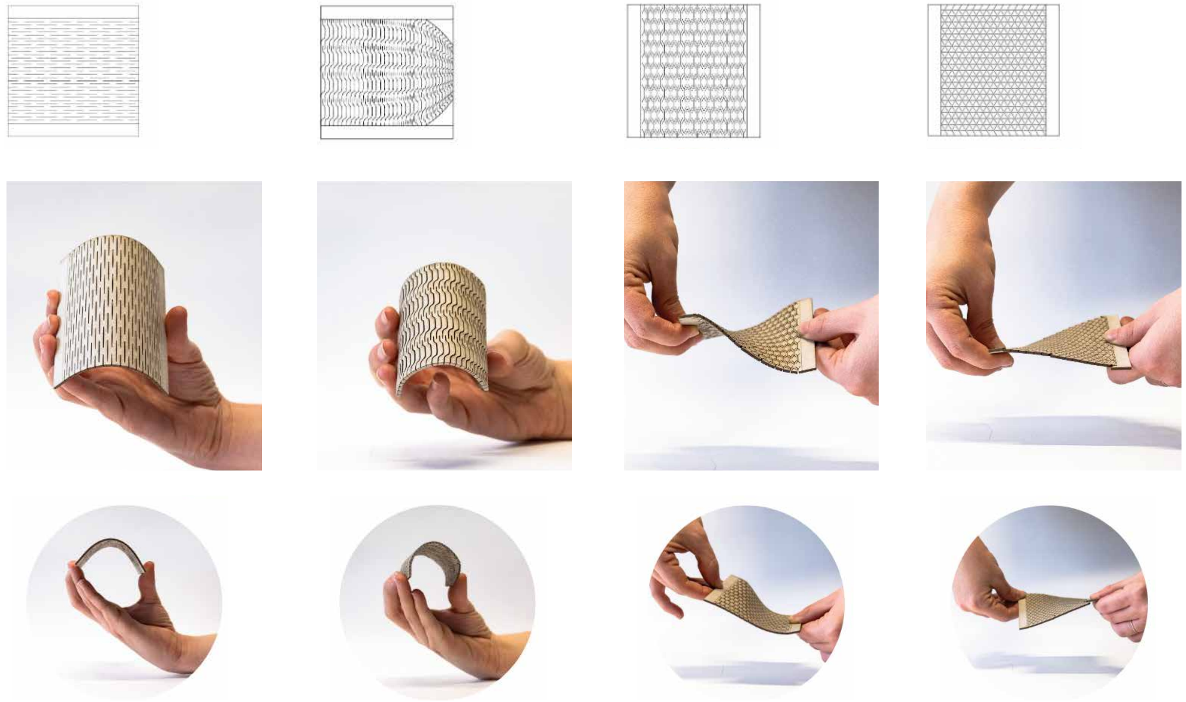
As part of an ITECH course focused on **applying biomimicry to parametric design**, we drew inspiration from weaver ants and abstracted their ability to strategically bite and fold leaves to facilitate weaving and nest formation.

To translate this behavior into a design strategy, we developed a **kerfing algorithm** (a technique that introduces patterned cuts into a material to increase its flexibility). Based on a desired final shape, the **algorithm defined the optimal pattern, including the precise position and size of each cut needed to achieve the target curvature.**

This approach allowed us to computationally control material deformation, bridging natural strategies and digital fabrication.

Fabrication process of the mock-up: weaving

Experimentation with different kerfing patterns



Fabrication process of the mock-up: CNC cutting